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Equivariant Kähler model for Fujiki's class

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Abstract
Let X be a compact complex manifold in Fujiki's class $\mathcal{C}$, i.e., admitting a big $(1, 1)$-class $[\alpha]$. Consider $\mathrm{Aut}(X)$ the group of biholomorphic automorphisms and $\mathrm{Aut}_{[\alpha]}(X)$ the subgroup of automorphisms preserving the class $[\alpha]$ via pullback. We show that X admits an $\mathrm{Aut}_{[\alpha]}(X)$-equivariant Kähler model: there is a bimeromorphic holomorphic map $\sigma : \widetilde{X} \to X$ from a Kähler manifold $\widetilde{X}$ such that $\mathrm{Aut}_{[\alpha]}(X)$ lifts holomorphically via σ. There are several applications. We show that $\mathrm{Aut}_{[\alpha]}(X)$ is a Lie group with only finitely many components. This generalizes an early result of Fujiki and Lieberman on the Kähler case. We also show that every torsion subgroup of $\mathrm{Aut}(X)$ is almost abelian, and $\mathrm{Aut}(X)$ is finite if it is a torsion group.

Keywords Fujiki's class $\mathcal{C}$ · Kähler manifold · Automorphism group · Lie group · Jordan property

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1 Introduction

Let X be a compact complex manifold in *Fujiki's class* $\mathcal{C}$, i.e., one of the following three equivalent assumptions is satisfied:

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(1) X is the meromorphic image of a compact Kähler manifold;
(2) X is bimeromorphic to a compact Kähler manifold;
(3) X admits a big $(1,1)$-class $[\alpha]$.

We refer to [11, Definition 1.1 and Lemma 1.1], [19, Chapter IV, Theorem 5] and [10, Theorem 0.7] for the equivalence and some properties of Fujiki's class $\mathcal{C}$.

It is a natural question that can we study the group of biholomorphic automorphisms $\mathrm{Aut}(X)$ via some well-chosen Kähler model of X? So this requires us to trace back to the construction of the Kähler model. Indeed, Demailly and Paun showed in the proof of [10, Theorem 3.4] that if a compact complex manifold X admits a big $(1,1)$-class $[\alpha]$, then there is a bimeromorphic holomorphic map $\sigma\colon X' \to X$ from a Kähler manifold X' obtained by a sequence of blowups along smooth centres determined by the ideal sheaf $\mathcal{J}$ corresponding to some Kähler current T with analytic singularities (cf. Definition 2.2) in $[\alpha]$. In general, $\mathrm{Aut}(X)$ does not lift via σ, for the first very simple reason that the class $[\alpha]$ may not be preserved by $\mathrm{Aut}(X)$. Then we focus ourselves on the subgroup

$$\mathrm{Aut}_{[\alpha]}(X) := \{g \in \mathrm{Aut}(X) \mid g^*[\alpha] = [\alpha]\}.$$

However, one still cannot expect the lifting of $\mathrm{Aut}_{[\alpha]}(X)$, for the second reason that the blown-up ideal sheaf $\mathcal{J}$ is not $\mathrm{Aut}_{[\alpha]}(X)$-invariant. Therefore, we need to find another Kähler model in a more natural way. Our idea is to consider the ideal sheaf generated by $g^*\mathcal{J}$ for all $g \in \mathrm{Aut}_{[\alpha]}(X)$.

The following is our main result. Note that the lift action of $\mathrm{Aut}_{[\alpha]}(X)$ on $\widetilde{X}$ is given by $g(-) = (\sigma^{-1} \circ g \circ \sigma)(-)$. By a holomorphic lifting we mean that the lift action is also holomorphic, i.e., the induced map $\mathrm{Aut}_{[\alpha]}(X) \times \widetilde{X} \to \widetilde{X}$ is holomorphic.

Theorem 1.1 *Let X be a compact complex manifold in Fujiki's class $\mathcal{C}$. For any big $(1,1)$-class $[\alpha]$ on X, there exists a bimeromorphic holomorphic map $\sigma\colon \widetilde{X} \to X$ from a Kähler manifold $\widetilde{X}$ such that $\mathrm{Aut}_{[\alpha]}(X)$ lifts holomorphically via σ.*

Remark 1.2 The bimeromorphic holomorphic map σ in Theorem 1.1 is indeed obtained by a sequence of $\mathrm{Aut}_{[\alpha]}(X)$-equivariant blowups along smooth centres.

Let X be a compact complex manifold. It is known that $\mathrm{Aut}(X)$ has the natural structure of a complex Lie group acting biholomorphically on X (cf. [9]). Denote by $\mathrm{Aut}_0(X)$ the connected component of $\mathrm{Aut}(X)$ containing the identity. Since it is connected, the pullback action of $\mathrm{Aut}_0(X)$ on the (discrete) lattice $H^2(X,\mathbb{Z})$ is trivial. When the $\partial\overline{\partial}$-lemma holds on X, we have the Hodge decomposition that $H^{1,1}(X,\mathbb{C})$ is a subspace of $H^2(X,\mathbb{C})$ and then $\mathrm{Aut}_0(X)$ acts also trivially on $H^{1,1}(X,\mathbb{R})$. Note that $\partial\overline{\partial}$-lemma holds when X admits a big $(1,1)$-class. So in this paper, our $\mathrm{Aut}_0(X)$ is always a subgroup of $\mathrm{Aut}_{[\alpha]}(X)$ for any big $(1,1)$-class $[\alpha] \in H^{1,1}(X,\mathbb{R})$. We refer to [7, Lemma (5.15) and Proposition (5.17)] and [11, Proposition 1.6 and Corollary 1.7] for the details.

When X is a Kähler manifold with a Kähler form α, Fujiki (cf. [11, Theorem 4.8]) and Lieberman (cf. [15, Proposition 2.2]), separately, proved that

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty.$$

Their proof heavily relies on the Kähler form α (or at least the existence of a Kähler form).

Nevertheless, with the help of our Theorem 1.1, we can show the following result.

Corollary 1.3 *Let X be a compact complex manifold in Fujiki's class $\mathcal{C}$. Then*

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty$$

for any big $(1,1)$-class $[\alpha]$ on X.

We also give applications on torsion group actions. When X is a projective variety defined over any field k of characteristic 0, Javanpeykar [12, Theorem 1.4] showed that the group of k-automorphisms $\mathrm{Aut}_k(X)$ is finite if it is torsion. We show that the same result holds true for normal compact complex spaces in Fujiki's class $\mathcal{C}$.

Corollary 1.4 *Let X be a normal compact complex space in Fujiki's class $\mathcal{C}$. Then $\mathrm{Aut}(X)/\mathrm{Aut}_0(X)$ has bounded torsion subgroups, i.e., there is a constant C such that $|G| \leq C$ for any torsion subgroup $G \leq \mathrm{Aut}(X)/\mathrm{Aut}_0(X)$. In particular, $\mathrm{Aut}(X)$ is a torsion group if and only if it is finite.*

In the previous work [16], Perroni, Zhang and the second author showed the Jordan property for $\mathrm{Aut}(X)$ when X is a compact complex space in Fujiki's class $\mathcal{C}$. The proof there developed a trick by finding some invariant Kähler submanifold Z (with $\mathrm{Aut}(X)$ shrunk a bit) and transferring the attention to a new compact Kähler manifold X': the compactified normal bundle $\mathbb{P}_Z(\mathcal{N}_{Z/X} \oplus \mathcal{O}_Z)$. Note however $\mathrm{Aut}(X')$ only keeps tracking of finite subgroups (more generally reductive subgroups or torsion subgroups) of $\mathrm{Aut}(X)$.

Now with the help of Corollary 1.3, we can provide an alternative proof.

Corollary 1.5 *Let X be a compact complex space in Fujiki's class $\mathcal{C}$. Then there exists a constant J such that any torsion subgroup G of $\mathrm{Aut}(X)$ has an abelian subgroup $H \leq G$ with $[G : H] \leq J$. In particular, $\mathrm{Aut}(X)$ has Jordan property.*

2 Preliminaries

Let X be a compact complex manifold with a fixed positive definite Hermitian form ω. Let α be a closed $(1,1)$-form. We use $[\alpha]$ to represent its class in $H^{1,1}(X, \mathbb{R})$. We define the following positivity notions (independent of the choice of ω):

- $[\alpha]$ is *Kähler* if it contains a Kähler form, i.e., if there is a smooth function φ such that $\alpha + \frac{\sqrt{-1}}{2\pi}\partial\overline{\partial}\varphi \geq \epsilon\omega$ on X for some $\epsilon > 0$.
- $[\alpha]$ is *big* if it contains a Kähler current T, i.e., if there is a quasi-plurisubharmonic function $\varphi\colon X \to \mathbb{R} \cup \{-\infty\}$ such that $T := \alpha + \frac{\sqrt{-1}}{2\pi}\partial\overline{\partial}\varphi \geq \epsilon\omega$ holds weakly as currents on X for some $\epsilon > 0$.

Recall that a *quasi-plurisubharmonic* function means that locally it is given by the sum of a plurisubharmonic function plus a smooth function.

We recall the definition of the non-Kähler locus of a big class $[\alpha]$.

Definition 2.1 Let X be a compact complex manifold and $[\alpha]$ a big $(1,1)$-class. Then the *non-Kähler locus* of $[\alpha]$ (or α) is defined and denoted by

$$E_{nK}(\alpha) := E_{nK}([\alpha]) := \bigcap_{T\in[\alpha]} \mathrm{Sing}(T),$$

where the intersection ranges over all Kähler currents $T = \alpha + \sqrt{-1}\partial\bar{\partial}\varphi$ in the class $[\alpha]$, and $\mathrm{Sing}(T)$ is the complement of the set of points $x \in X$ such that φ is smooth at x.

We recall the basic definition of analytic singularities (cf. [3, Section 2.1]). Note that the data $(\mathcal{J}, c)$ below is not uniquely determined by the function φ with analytic singularities.

Definition 2.2 Let X be a compact complex manifold and $[\alpha]$ a closed $(1,1)$-class.

(1) Given a coherent ideal sheaf $\mathcal{J}$ and a constant $c > 0$, we say that a function φ has *singularities of type* $(\mathcal{J}, c)$ if locally it can be written as

$$\varphi = c\log\Big(\sum_{j=1}^{n}|f_j|^2\Big) + h$$

for some local generators (f_j) of $\mathcal{J}$ and some smooth function h.

(2) We say that φ has *analytic singularities* if it has singularities of type $(\mathcal{J}, c)$ for some coherent ideal sheaf $\mathcal{J}$ and some constant $c > 0$.

(3) We also say that a closed current $T \in [\alpha]$ has *analytic singularities* if it can be written as $T = \alpha + \frac{\sqrt{-1}}{2\pi}\partial\bar{\partial}\varphi$ such that the potential function φ has analytic singularities.

The following is a direct application of the regularization theorem by Demailly [5]; see also [10, Theorem 3.2].

Theorem 2.3 *Let X be a compact complex manifold with a big $(1,1)$-class $[\alpha]$. Then there exists a Kähler current $T \in [\alpha]$ with analytic singularities.*

Boucksom [4, Theorem 3.17] observed further that indeed one can find a Kähler current with analytic singularities and also with minimal singular locus.

Theorem 2.4 *Let X be a compact complex manifold with a big $(1,1)$-class $[\alpha]$. Then there exists a Kähler current $T \in [\alpha]$ with analytic singularities such that $\mathrm{Sing}(T) = E_{nK}(\alpha)$. In particular, $E_{nK}(\alpha)$ is a closed analytic subspace of X and $E_{nK}(\alpha) = \emptyset$ if and only if $[\alpha]$ is Kähler.*

3 Equivariant Kähler Model

In this section, we prove Theorem 1.1.

First, we recall the following theorem on the equivariant log resolution of an ideal sheaf by Bierstone and Milman [2, Theorem 1.10].

Theorem 3.1 *Let X be a compact complex manifold with a coherent ideal sheaf $\mathcal{J}$. Let $G \leq \mathrm{Aut}(X)$ such that $g^*\mathcal{J} = \mathcal{J}$ for any $g \in G$. Then there is a finite sequence*

$$X_k \xrightarrow{\sigma_k} \cdots \xrightarrow{\sigma_2} X_1 \xrightarrow{\sigma_1} X_0 = X$$

of G-equivariant blowups σ_j, $j = 1, \ldots, k$, along smooth centres, such that the inverse image ideal sheaf $\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{X_k}$ is a normal-crossings divisor.

Proof In [2, Theorem 1.10], the sequence is taken as blowups along $\mathrm{inv}_{\mathcal{J}}$-admissible centres. Note that an $\mathrm{inv}_{\mathcal{J}}$-admissible centre is determined by $\mathcal{J}$ itself and hence G-invariant. So the blowups are G-equivariant. We refer to [20] for the details. □

The following lemma is well-known, and we give a proof for the convenience of the readers. The result holds true in the algebraic setting with the same proof.

Lemma 3.2 *Let $\sigma : X' \to X$ be a bimeromorphic holomorphic map of compact complex spaces. Let G be a complex Lie group acting holomorphically on X such that G lifts via σ (i.e., each $g \in G$ lifts to an automorphism of X'). Then the induced action of G on X' is also holomorphic.*

Proof Consider the holomorphic map

$$\phi \colon G \times X' \times X' \to G \times X \times X$$

via $(g, x', y') \mapsto (g, \sigma(x'), \sigma(y'))$.

Consider the graph of G on X:

$$\Gamma := \{(g, x, y) \in G \times X \times X \mid y = g(x)\}$$

which is a closed analytic subspace of $G \times X \times X$ since the G-action on X is holomorphic.

For each $g \in G$, let $\Gamma_{g|_X}$ be the graph of $g|_X$ viewed as a fibre of $\Gamma \to G$ over g. Note that $g|_{X'} \colon X' \to X'$ is an automorphism, so we can define $\Gamma_{g|_{X'}}$ similarly. Then $\Gamma_{g|_{X'}}$ is the proper transform of $\Gamma_{g|_X}$ in $G \times X' \times X'$ via ϕ. Let Γ' be the proper transform of Γ in $G \times X' \times X'$. Then Γ' is a closed analytic subspace of $G \times X' \times X'$. By taking the first projection, the fibre of $\Gamma' \to G$ over g is just $\Gamma_{g|_{X'}} \cong X'$ since $g \colon X' \to X'$ is biholomorphic. Therefore, the projection to the first two factors $\Gamma' \to G \times X'$ is biholomorphic. Note that Γ' is the graph of G on X'. So the G-action on X' is holomorphic. □

Proof of Theorem 1.1 By Theorem 2.3, there exists a Kähler current

$$T = \alpha + \frac{\sqrt{-1}}{2\pi}\partial\bar{\partial}\varphi \in [\alpha]$$

with analytic singularities of type $(\mathcal{J}_\varphi, c)$ for some coherent ideal sheaf $\mathcal{J}_\varphi$ and $c > 0$. This means that (locally) we may write

$$\varphi = c \log \Big(\sum_{j=1}^{n} |f_j|^2 \Big) + h$$

where (f_j) are local generators of $\mathcal{J}_\varphi$ and h is smooth. Consider the $\mathrm{Aut}_{[\alpha]}(X)$-invariant ideal sheaf

$$\mathcal{J} := \sum_{g \in \mathrm{Aut}_{[\alpha]}(X)} g^* \mathcal{J}_\varphi.$$

Since X is compact, $\mathcal{J}$ is also coherent (cf. [6, Chapter II, Property (3.22)]). Then we may write

$$\mathcal{J} = \sum_{i=1}^{m} \mathcal{J}_i$$

where $\mathcal{J}_i := g_i^* \mathcal{J}_\varphi$ for some $g_i \in \mathrm{Aut}_{[\alpha]}(X)$.

By Theorem 3.1, there is an $\mathrm{Aut}_{[\alpha]}(X)$-equivariant biholomorphic holomorphic map

$$\sigma : \widetilde{X} \to X$$

which is obtained by a sequence of blowups along smooth centres such that $\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}$ is invertible.

Let $\mathcal{J}_i' := \sigma^{-1}\mathcal{J}_i \cdot \mathcal{O}_{\widetilde{X}}$. Then

$$\sum_{i=1}^{m} \mathcal{J}_i' = \sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}$$

is invertible.

Denote by

$$\widetilde{\mathcal{J}_i} := \mathcal{J}_i' \cdot \big(\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}\big)^{-1}$$

which is still an ideal sheaf. Then

$$\sum_{i=1}^{m} \widetilde{\mathcal{J}_i} = \big(\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}\big) \cdot \big(\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}\big)^{-1} = \mathcal{O}_{\widetilde{X}}.$$

We give several more notations.

- Let s be the local generator of invertible sheaf $\sigma^{-1}\mathcal{J} \cdot \mathcal{O}_{\widetilde{X}}$.

- We define a positive current D which is the integration current along the divisor $\{s = 0\}$. By the Poincaré-Lelong formula, D can be locally written as $\frac{\sqrt{-1}}{2\pi}\partial\overline{\partial}\log|s|^2$.
- Fix a (positive definite) Hermitian form ω on X such that $g_i^*T \geq \omega$ for each $1 \leq i \leq m$.
- Let $E_1, \cdots, E_r$ be the (full) exceptional prime divisors of σ.

Claim 3.3 *One can find $\delta_1, \cdots, \delta_r > 0$ small enough and a closed smooth $(1,1)$-form u_E in the class $[E := \delta_1 E_1 + \cdots + \delta_r E_r]$ such that $\sigma^*\omega - u_E$ is a positive definite Hermitian form.*

Proof of Claim 3.3 It suffices for us to consider the case when σ is a one-step blowup. Then a proof as [10, Proof of Lemma 3.5] also works here. □

We show that $\widetilde{X}$ is Kähler by the following claim.

Claim 3.4 *The class $[\widetilde{\alpha}] := [\sigma^*\alpha - cD - u_E]$ contains a Kähler form.*

Let $T_i' := \sigma^* g_i^* T$ and $f_{j,i}' := f_j \circ g_i \circ \sigma$. Then the potential function of T_i' is locally of the form

$$\varphi_i' = c\log\Big(\sum_{j=1}^{n}|f_{j,i}'|^2\Big) + h \circ g_i \circ \sigma$$

and $(f_{j,i}')_j$ are local generators of $\mathcal{J}_i'$. Let s_i be the greatest common divisor of the $(f_{j,i}')$'s. Then we can write down the Siu's decomposition

$$T_i' = R_i + cD_i$$

where D_i is the integration current along the divisor $\{s_i = 0\}$ which can be locally written as $\frac{\sqrt{-1}}{2\pi}\partial\overline{\partial}\log|s_i|^2$ and $R_i \geq \sigma^*\omega$ such that the Lelong super-level (analytic) set $E_c(R_i)$ has codimension at least 2 (cf. [4, Sections 2.2.1–2.2.2] and [3, Section 2.2]). Note that s is a factor of s_i since $\mathcal{J}_i' \subseteq \sigma^{-1}\mathcal{J}\cdot\mathcal{O}_{\widetilde{X}}$. Then we have $D_i \geq D$ and hence

$$T_i' - cD \geq R_i \geq \sigma^*\omega.$$

We now construct

$$\widetilde{T}_i := T_i' - cD - u_E = (\sigma^* g_i^*\alpha - u_E) + \frac{\sqrt{-1}}{2\pi}\partial\overline{\partial}\varphi_i' - cD \in [\widetilde{\alpha}].$$

By the construction, the potential function of $\widetilde{T}_i$ can be locally written as

$$\begin{aligned}\widetilde{\varphi}_i &= c\log\Big(\sum_{j=1}^{n}|f_{j,i}'|^2\Big) + h\circ g_i\circ\sigma - c\log|s|^2\\ &= c\log\Big(\sum_{j=1}^{n}|\widetilde{f}_{j,i}|^2\Big) + h\circ g_i\circ\sigma\end{aligned}$$

where $f'_{j,i} = \widetilde{f}_{j,i} \cdot s$. Note that

$$\widetilde{T}_i \geq \sigma^* \omega - u_E$$

while the right-hand side is positive definite by the construction before Claim 3.4. Therefore, $\widetilde{T}_i$ is also a Kähler current.

Note that $\widetilde{\mathcal{J}}_i$ is locally generated by $(\widetilde{f}_{j,i})_j$. Then

$$\bigcap_{i=1}^{m} \mathrm{Sing}(\widetilde{T}_i) = \bigcap_{i=1}^{m} V(\widetilde{\mathcal{J}}_i) = V\Big(\sum_{i=1}^{m} \widetilde{\mathcal{J}}_i\Big) = \emptyset$$

where $V(-)$ is the zeros of the ideal sheaf. In particular, $E_{nK}(\widetilde{\alpha}) = \emptyset$ and the claim is proved by Theorem 2.4.

Finally, note that the induced action of $\mathrm{Aut}_{[\alpha]}(X)$ on $\widetilde{X}$ is holomorphic by Lemma 3.2. □

4 Boundedness on Group Components

In this section, we are going to prove Corollaries 1.3 and 1.5. Our Corollary 1.3 plays a key role in the reduction to the Kähler case.

Recall that Fujiki [11, Theorem 4.8] and Lieberman [15, Proposition 2.2] and showed separately that

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty$$

for a compact Kähler manifold X and a Kähler form α. For the convenience of our proof for Corollary 1.3, we give a version on a big class generalized by Dinh, Hu and Zhang; see [8, Theorem 2.1] for a more generalized setting. Note however their proof still requires the existence of a Kähler form. We refer to [17, Propositions 2.9 and 3.6] for a generalized explanation by cone analysis and linear algebra.

Theorem 4.1 *Let X be a compact Kähler manifold. Then*

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty$$

for any big $(1,1)$-class $[\alpha]$.

We need the following lemma concerning the descending of connected complex Lie group action via a bimeromorphic holomorphic map.

Lemma 4.2 *Let $\sigma : X' \to X$ be a bimeromorphic holomorphic map of normal compact complex spaces. Then the map*

$$\tau : \mathrm{Aut}_0(X') \to \mathrm{Aut}_0(X), \quad g \mapsto \sigma \circ g \circ \sigma^{-1}$$

is an injective complex Lie group homomorphism.

Proof Note that $\sigma_*\mathcal{O}_{X'} = \mathcal{O}_X$ and $\mathrm{Aut}_0(X')$ is connected. Then the lemma is essentially a corollary of the rigidity lemma (cf. [1, § 2.4, Lemmas 1 and 2]). Note that τ is injective because σ is bimeromorphic. □

Theorem 4.3 *Let X be a normal compact complex space in Fujiki's class $\mathcal{C}$. Then there is a Kähler model $\sigma : \widetilde{X} \to X$ such that the map*

$$\tau : \mathrm{Aut}_0(\widetilde{X}) \to \mathrm{Aut}_0(X)$$

via $\tau(g) := \sigma \circ g \circ \sigma^{-1}$ is a complex Lie group isomorphism.

Proof Let $\pi : X' \to X$ be an $\mathrm{Aut}(X)$-equivariant resolution of singularities (cf. [2, Theorem 13.2]). It follows from Lemmas 3.2 and 4.2 that the map

$$\mathrm{Aut}_0(X') \to \mathrm{Aut}_0(X), \quad g \mapsto \pi \circ g \circ \pi^{-1}$$

is a complex Lie group isomorphism since X is normal (see also [11, Lemma 2.5]). Note that X' is also in Fujiki's class $\mathcal{C}$. So we may replace X with X' and assume that X is smooth.

Since $\mathrm{Aut}_0(X)$ is connected, its pullback action on $H^2(X, \mathbb{Z})$ is trivial. Note that the $\partial\bar{\partial}$-lemma holds for compact complex manifolds in Fujiki's class $\mathcal{C}$. So $H^{1,1}(X, \mathbb{R})$ is a subspace of $H^2(X, \mathbb{R})$ and $\mathrm{Aut}_0(X)$ acts trivially on $H^{1,1}(X, \mathbb{R})$. Let $[\alpha] \in H^{1,1}(X, \mathbb{R})$ be a big $(1, 1)$-class. Note that $\mathrm{Aut}_0(X) \leq \mathrm{Aut}_{[\alpha]}(X)$.

We take σ as in Theorem 1.1. Then the map

$$\phi : \mathrm{Aut}_0(X) \times \widetilde{X} \to \widetilde{X}$$

via $(g, \widetilde{x}) \mapsto (\sigma^{-1} \circ g \circ \sigma)(\widetilde{x})$ is well-defined and holomorphic. In particular, $\mathrm{Aut}_0(X)$ lifts to a (unique) subgroup of $\mathrm{Aut}_0(\widetilde{X})$. By Lemma 4.2, the map

$$\tau : \mathrm{Aut}_0(\widetilde{X}) \to \mathrm{Aut}_0(X)$$

via $\tau(g) := \sigma \circ g \circ \sigma^{-1}$ is an injective complex Lie group homomorphism. We just see the surjectivity of τ by the lifting property. So τ is isomorphic. □

Proof of Corollary 1.3 Let $[\alpha]$ be a big $(1, 1)$-class. By Theorem 1.1, there is a Kähler model $\sigma : \widetilde{X} \to X$ such that $\mathrm{Aut}_{[\alpha]}(X)$ lifts to a group $G \leq \mathrm{Aut}(\widetilde{X})$ via σ. By Theorem 4.3, $\mathrm{Aut}_0(\widetilde{X}) \leq G$. Note that $G \leq \mathrm{Aut}_{\sigma^*[\alpha]}(\widetilde{X})$ and $\sigma^*[\alpha]$ is still big. Since $\widetilde{X}$ is Kähler, we have that

$$[\mathrm{Aut}_{\sigma^*[\alpha]}(\widetilde{X}) : \mathrm{Aut}_0(\widetilde{X})] < \infty$$

by Theorem 4.1. Finally, note that

$$\mathrm{Aut}_{[\alpha]}(X)/\mathrm{Aut}_0(X) \cong G/\mathrm{Aut}_0(\widetilde{X}) \leq \mathrm{Aut}_{\sigma^*[\alpha]}(\widetilde{X})/\mathrm{Aut}_0(\widetilde{X}).$$

So the corollary is proved. □

Remark 4.4 Lemma 4.2 and also Theorem 4.3 fail in general if X has non-normal singularities such that $\sigma_*\mathcal{O}_{X'} = \mathcal{O}_X$ does not hold. A simple example is by taking $X' = \mathbb{P}^1$, $X = \{y^2z - x^3 = 0\}$ the cuspidal curve, and σ just the normalization of X. Note that $\mathrm{Aut}(X') = \mathrm{Aut}_0(X') = \mathrm{PGL}(2)$ while $\mathrm{Aut}_0(X)$ is (conjugate to) the subgroup of upper triangular matrices in $\mathrm{PGL}(2)$.

5 Torsion Group Actions

In this section, we will prove Corollaries 1.4 and 1.5. Our Theorem 1.1 plays a key role in the reduction to the connected Lie group action.

The following result holds true for algebraic groups defined over an algebraically closed field of characteristic 0; see [12, Lemma 5.4]. In the case of connected real Lie groups, the proof is simple by considering the one parameter subgroups.

Lemma 5.1 *A torsion connected real Lie group G is trivial.*

Proof Note that a one parameter subgroup is either $\mathbb{R}$ or S^1. So if G is positive dimensional, G contains an element of infinite order. □

For general linear groups over number fields, we may even have boundedness on their torsion subgroups.

Lemma 5.2 *Let G be a torsion subgroup of $\mathrm{GL}_n(K)$ where K is a number field. Then $|G| \leq N$ for some constant N depending only on n and K.*

Proof By the Minkowski's theorem (cf. [18, Theorem 5, and § 4.3]), there is a constant M depending only on n and K such that the order $o(g) \leq M$ for any $g \in \mathrm{GL}_n(K)$ with finite order. By the Burnside's first theorem (cf. [13, Chapter 3, § 9, Theorem (9.4)]), $|G| \leq N := M^{n^3}$ is finite. □

Proof of Corollary 1.4 Let G be a torsion subgroup of $\mathrm{Aut}(X)/\mathrm{Aut}_0(X)$. Let $\pi : X' \to X$ be an $\mathrm{Aut}(X)$-equivariant resolution of singularities (see also [2, Theorem 13.2]), with $\mathrm{Aut}(X)$ lifts to a (unique) subgroup of $\mathrm{Aut}(X')$ via π. Note that $\mathrm{Aut}_0(X)$ lifts (isomorphically) to $\mathrm{Aut}_0(X')$ (cf. [11, Lemma 2.5]). Then G also lifts to a (unique) torsion subgroup of $\mathrm{Aut}(X')/\mathrm{Aut}_0(X')$. Note that X' is also in Fujiki's class $\mathcal{C}$. Therefore, we may replace X with X' and assume that X is smooth.

Since the pullback action $\mathrm{Aut}_0(X)|_{H^2(X,\mathbb{Q})}$ is trivial, we have the following exact sequence

$$1 \longrightarrow G_\tau \longrightarrow G \longrightarrow G|_{H^2(X,\mathbb{Q})} \longrightarrow 1$$

where G_τ is the kernel. Note that $G|_{H^2(X,\mathbb{Q})}$ is torsion and hence finite with order bounded by some N depending only on $H^2(X,\mathbb{Q})$ (and hence only on X) by Lemma 5.2. Denote by

$$\mathrm{Aut}_\tau(X) := \{g \in \mathrm{Aut}(X) \mid g^*|_{H^2(X,\mathbb{Q})} = \mathrm{id}\}.$$

Note that

$$G_\tau \leq \mathrm{Aut}_\tau(X)/\mathrm{Aut}_0(X) \leq \mathrm{Aut}_{[\alpha]}(X)/\mathrm{Aut}_0(X)$$

for any big $(1,1)$-class $[\alpha]$. By Corollary 1.3, we have

$$|G_\tau| \leq C := [\mathrm{Aut}_\tau(X) : \mathrm{Aut}_0(X)],$$

where C depends only on X. Then $|G| \leq N \cdot C$ and we get an upper bound.

Finally, if $\mathrm{Aut}(X)$ is torsion, then $\mathrm{Aut}_0(X)$ is trivial by Lemma 5.1 and hence $\mathrm{Aut}(X)$ is finite. □

Remark 5.3 Currently, the normality assumption on X is required in the proof of Corollary 1.4. The reason is that the equivariant resolution of singularities may enlarge $\mathrm{Aut}_0(X)$; see Remark 4.4.

It is well known that any torsion subgroup of the general linear group is almost abelian by the Jordan–Schur lemma. Lee generalized it to the case of connected real Lie groups; see [14].

Theorem 5.4 *Let G be a connected real Lie group. Then any torsion subgroup $H \leq G$ has an abelian subgroup $H' \leq H$ with $[H : H'] \leq J$ where J is a constant depending only on G.*

Proof of Corollary 1.5 First, by taking an $\mathrm{Aut}(X)$-equivariant resolution of singularities (cf. [2, Theorem 13.2]) which is still in Fujiki's class $\mathcal{C}$, we may assume X is smooth.

Let $G \leq \mathrm{Aut}(X)$ be a torsion subgroup. Denote by

$$\mathrm{Aut}_\tau(X) := \{g \in \mathrm{Aut}(X) \mid g^*|_{H^2(X,\mathbb{Q})} = \mathrm{id}\}.$$

We may view $G/G\cap\mathrm{Aut}_\tau(X)$ as a torsion subgroup of $\mathrm{GL}(H^2(X,\mathbb{Q}))$. By Lemma 5.2, we have

$$[G : G \cap \mathrm{Aut}_\tau(X)] \leq N$$

for some constant N depending only on $H^2(X,\mathbb{Q})$ (and hence only on X).

Since $\mathrm{Aut}_\tau(X) \subseteq \mathrm{Aut}_{[\alpha]}(X)$ and by Corollary 1.3, we see that

$$C := [\mathrm{Aut}_\tau(X) : \mathrm{Aut}_0(X)] < \infty$$

and hence

$$[G : G \cap \mathrm{Aut}_0(X)] \leq N \cdot C.$$

By Theorem 5.4, there is an abelian subgroup $H \leq G \cap \mathrm{Aut}_0(X)$ such that

$$[G \cap \mathrm{Aut}_0(X) : H] \leq J_0$$

where J_0 is a constant depending only on $\mathrm{Aut}_0(X)$ (and hence only on X).

Together, we have

$$[G : H] \leq J := N \cdot C \cdot J_0$$

as desired. □

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